HCL AI Employee Assistant - Coding Standards

Version: 2026.1
Department: Engineering

1. Coding Principles

Write clean, readable, and maintainable code. Avoid unnecessary complexity and always follow established coding guidelines.

2. Naming Conventions

Use meaningful variable and function names. Classes should follow PascalCase. Variables and functions should follow snake_case for Python projects. Constants should be written in UPPER_CASE.

3. Git Workflow

Every new feature should be developed in a separate feature branch. Commit messages should clearly describe the change. Pull Requests (PRs) must be created before merging into the main branch.

4. Code Review

Every Pull Request must be reviewed by at least one team member before merging. Reviewers should verify functionality, readability, security, and coding standards.

5. Documentation

Public functions and classes should include proper documentation. Complex logic should be explained with concise comments when necessary. Project README files must remain up to date.

6. Testing

Developers should write unit tests for new features whenever possible. Critical bugs must include regression tests before deployment.

7. Logging & Error Handling

Applications should use structured logging instead of print statements. Exceptions should be handled gracefully and meaningful error messages should be logged. Sensitive information such as passwords or API keys must never be logged.

8. Security Best Practices

Do not hardcode API keys, passwords, or secrets in source code. Use environment variables and secure secret management solutions. Validate all user inputs to reduce security risks.

9. Important Notes

Following coding standards improves software quality, maintainability, team collaboration, and long-term project success.